

IAS PRELIMS/MAINS

Ecology & Environment

A stylized illustration of a green city skyline with various skyscrapers, wind turbines, and a sun, set against a light green background with a globe-like shape.

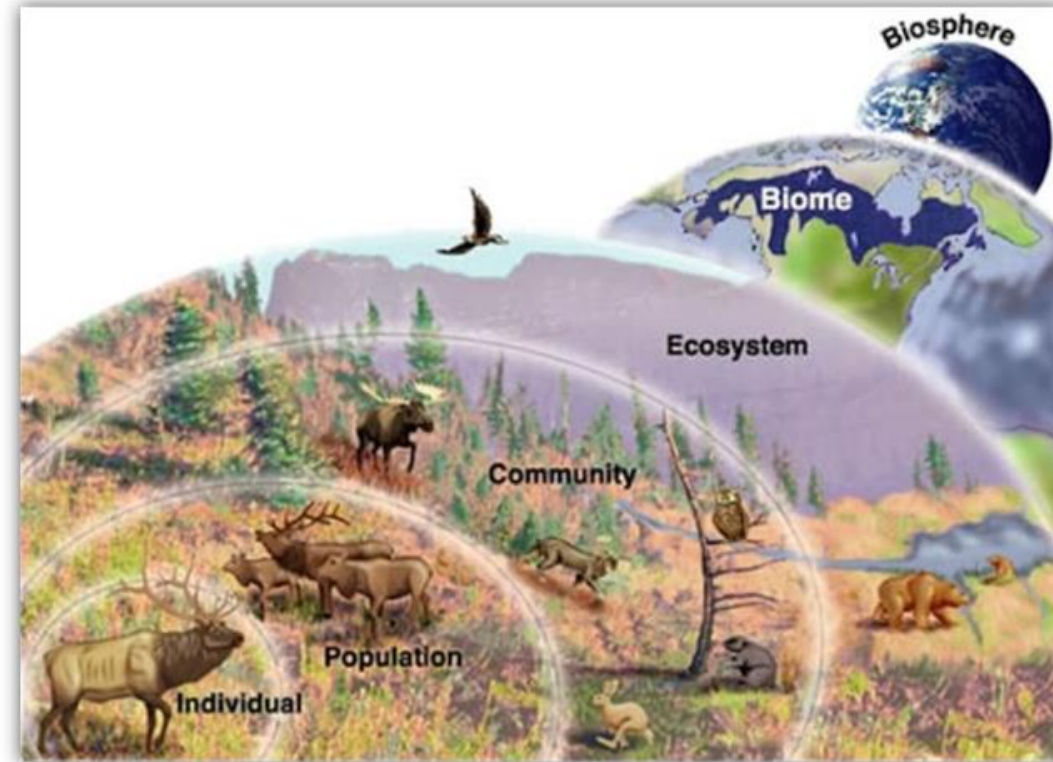
A COMPREHENSIVE COVERAGE

MODULE - 3

BASIC CONCEPTS - 3

ECOLOGY

- Ecology is the branch of study concerned with the relations of organisms to one another and to their physical surroundings (environment).
- Ecology- Etymology is oikos+logy. Oikos means house or dwelling.



ECOLOGY

Thus, there are 3 things that we study in ecology

- Living organisms
- Their relationship & interactions with other organisms.
E.g. what kind of relationship is there between the bacteria rhizobium and plant roots, or competition among herbivorous animals for food etc.
- Their interactions with surrounding environment. It means how the organism responds to climate, soil, topography, water availability, etc.

Example:

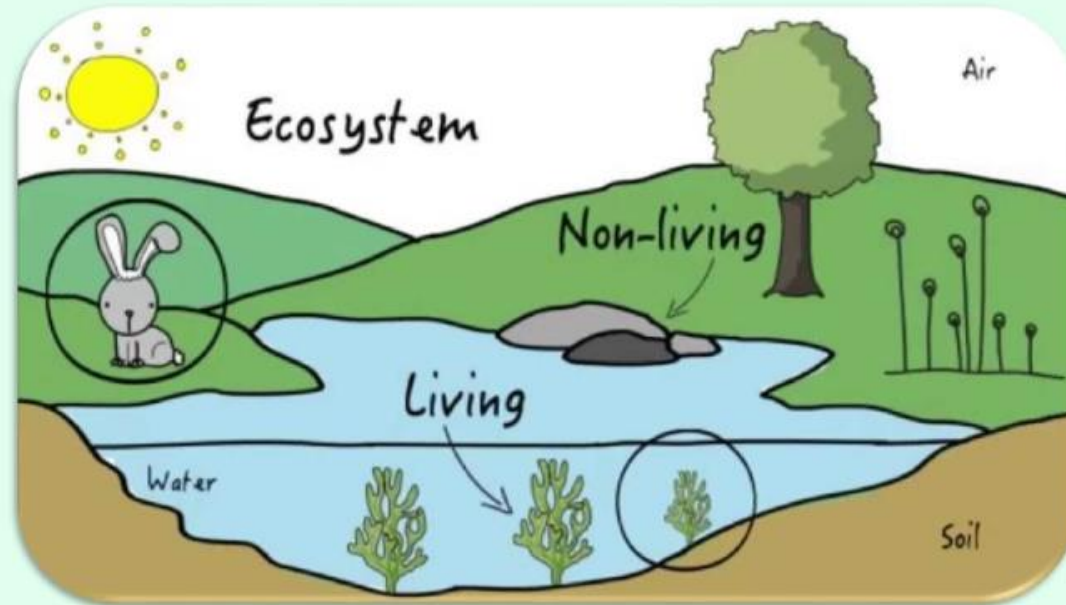
Animals in deserts have long ears to increase heat loss and keep themselves cool while animals in cold areas have small ears to prevent heat loss.



The interactions with other organisms and environment include Energy Flow (Food chains/Food web) and Nutrient Cycles.

THE MEANING OF ENVIRONMENT AND ECOLOGY

- An Ecosystem is the functional unit of nature (like how cell is the functional unit of life).
- It includes environment (biotic+abiotic) and ecology (how living organisms interact with each other and with the environment).
- Thus $\text{Ecosystem} = \text{Ecology} + \text{Environment}$



LEVELS OF ORGANIZATIONS IN ECOLOGY

Ecology not only deals with the study of the relationship of individual organisms with their environment, but also with the study of populations, communities, ecosystems, biomes, and biosphere as a whole.

1 Organism: It is an individual living being that has the ability to act or function independently. It may be any organism.

2 Species: Species are a group of similar organisms consisting of similar genetic structure and are capable of interbreeding.

3 Population: Population is large number of organisms which belong to one species.
E.g.: populations of tigers, lions, rhinos etc.

4 Community: Populations of two or more different species living together in an area is known as community.

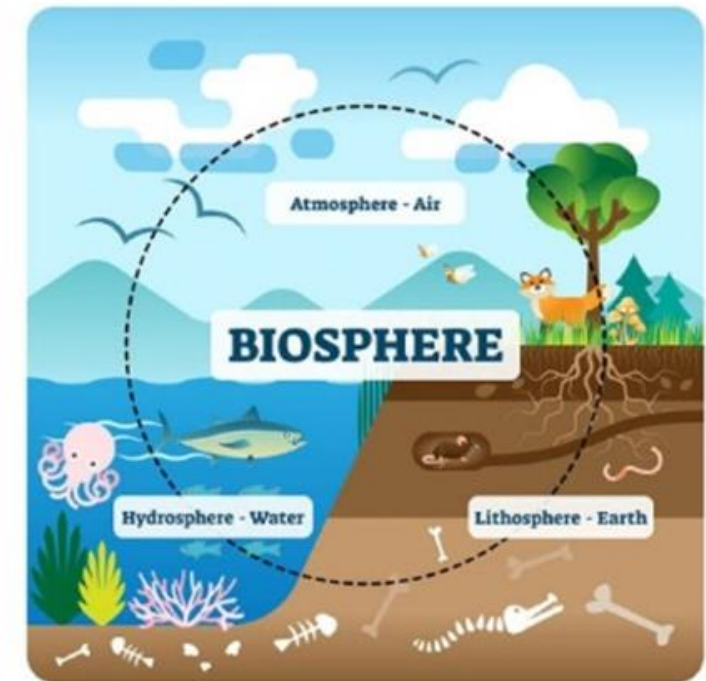


Ecology not only deals with the study of the relationship of individual organisms with their environment, but also with the study of populations, communities, ecosystems, biomes, and biosphere as a whole.

5 Ecosystem: An ecosystem is a community of organisms interacting with each other and with their environment. The interaction leads to flow of energy and cycling of nutrients.

6 Biome: Biome is a large naturally occurring community of flora and fauna occupying a major habitat.
E.g. Rainforest biome or tundra biome. Plants and animals in a biome have common characteristics due to similar climatic condition throughout the biome.

7 Biosphere: The biosphere is the biological component of earth which includes the lithosphere, hydrosphere and atmosphere. The biosphere includes all living organisms on earth, together with the dead organic matter produced by them.

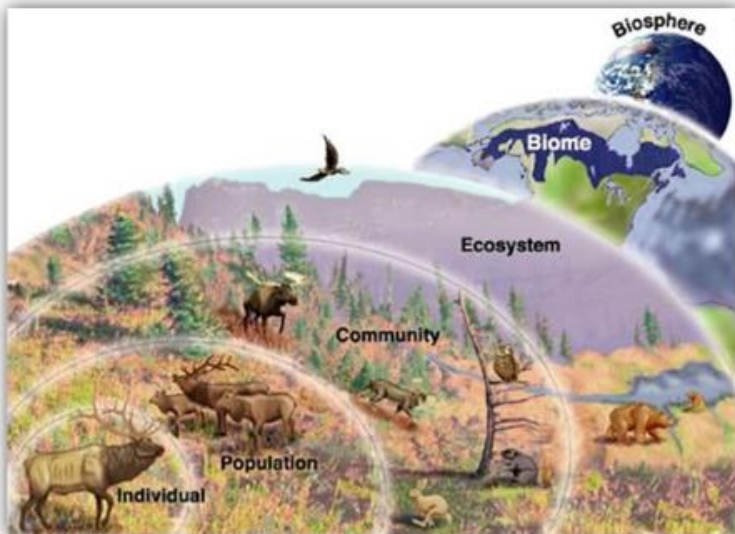


LEVELS OF ORGANIZATION

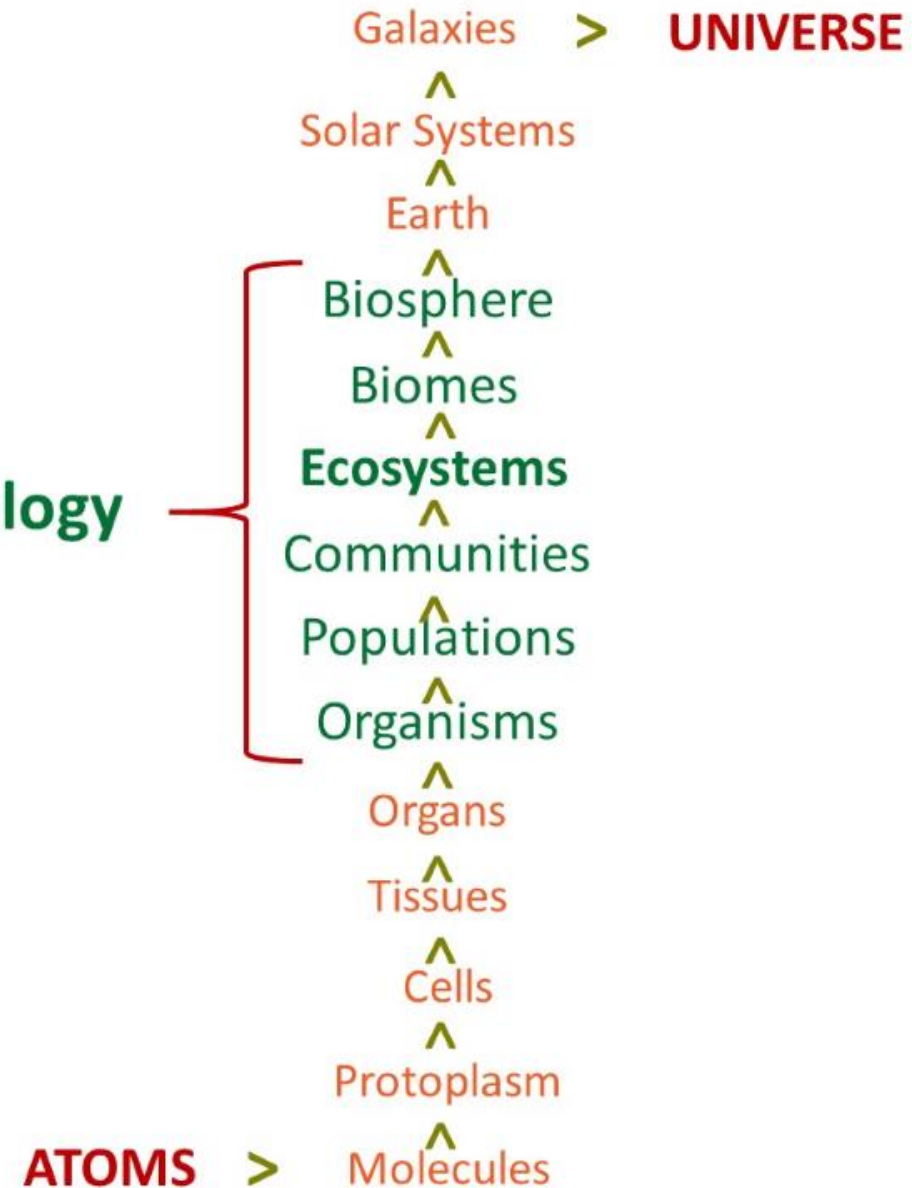
Ecologist study

Organisms ranging from the various levels of organization

- Species
- Population
- Community
- Ecosystem
- Biome
- Biosphere



Ecology



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